



'Magic' material that enables development of quantum electronic devices



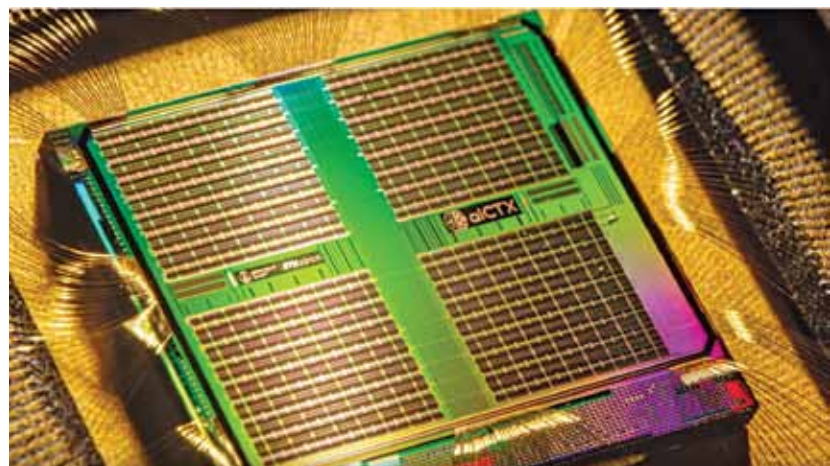
MIT graduate student Daniel Rodan-Legrain holds a chip carrier used in his work on developing new graphene-based electronic devices. He stands next to a dilution refrigerator similar to that used in the work (Credit: <https://news.mit.edu>)

MIT researchers and colleagues have turned a 'magic' material based on graphene, which is composed of atomically thin layers of carbon, into useful electronic devices. Normally, such devices are key to the quantum electronics industry. The approach solves a variety of problems associated with more complicated processes. As a result, the work could usher in a new generation of quantum electronic devices for applications including quantum computing. Further, the devices can be superconducting or conduct electricity without resistance. The overall 'magic' material, formally known as magic-angle twisted bilayer graphene (MATBG), has inspired a new field known as twistronics. The developed electronic devices include a Josephson junction or superconducting switch, a spectroscopic tunnelling device, and a single-electron transistor.



Neuromorphic chip that precisely detects high-frequency oscillations

Current neural network algorithms help us solve an incredible number of problems. However, the electronic devices used to run these algorithms still require too much processing power. These artificial intelligence (AI) systems simply cannot compete with an actual brain when it comes to processing sensory information or interactions with the environment in real time. Now, an interdisciplinary research team at the University of Zurich, the ETH Zurich, and the University Hospital Zurich have developed a chip based on neuromorphic technology that reliably and accurately recognises complex biosignals. The scientists were able to use this technology to successfully detect previously recorded high-frequency oscillations. These specific waves, measured using an intracranial electroencephalogram (iEEG), have proven to be promising biomarkers for identifying the brain tissue that causes epileptic seizures.



The neuromorphic chip reliably and precisely detects high-frequency oscillations in previously recorded intracranial EEG (Credit: www.media.uzh.ch/en)